

**AIMS**King Abdulaziz University Jeddah, Saudi Arabia
Faculty of Computing and Information Technology**Test 2****CPCS-214 Section: CA****Computer Organization and Architecture (I)**
Spring 2012-2013 (Second Semester 1433 - 1434)**Coordinator Name:** Dr. Abdulla A. Basuhail, Assistant Professor**Teacher Name:** Dr. Abdulla A. Basuhail, Assistant Professor

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SN

13

	Max Marks	Obtained Marks
SO "A" Score	13	$8\frac{1}{2} + 1$
SO "B" Score	17	17
SO "C" Score	10	10
SO "Other" Score	0	

Total Score:(Optional, System
will auto total)

40

 $35\frac{1}{2} + 1$ $36\frac{1}{2}$

1. Represent the binary number: 1100 using 8-bit, if the number is:

a. Signed.

CS-B: ²3 Points

²2

11111100

b. Unsigned.

CS-B: ²3 Points

²2

00001100

c. Show by verification which representation (a or b) is equivalent to the original number

(1100)₂

CS-C: ²3 Points

²3

if this number is in unsigned representation

then the equivalent no. is b ✓

because $b = (12)_{10}$ and $a = (-4)_{10}$ after taking the two's complement and the original no. is equal to $(12)_{10}$

if that no. $(1100)_2$ is represented in sign-and-magnitude

then it's equal to a

because $b = (12)_{10}$ and $a = (-4)_{10}$ where the original no. is equal to $(-4)_{10}$ after taking the two's complement

2. The contents of register $\$s0 = 12000012_{\text{hex}}$ and register $\$s1 = 80000012_{\text{hex}}$. What is the content of register $\$t0$ after executing the following instructions (Illustrate your answer).

sltu $\$t0, \$s0, \$s1$

CS-C: 4 Points

5

$$12000012_{\text{hex}} = (0001\ 0010\ 0000\ 0000\ 0000\ 0000\ 0001\ 0010)_2$$

$$80000012_{\text{hex}} = (1000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0001\ 0010)_2$$

if we compared the hex number in $\$s0$ we will find that it's smaller than the number in $\$s1$

so $\$s0 < \$s1$ is true which $(1)_2$ in binary

so the content of $\$t0$ is $(1)_2$ or $(00000001)_{\text{hex}}$

3. Write MIPS code to add two single precision floating point numbers (Assume the numbers are already loaded in floating point registers).

CS-B: 4 Points

4

add.s \$f1, \$f2, \$f3

✓

4. Find the value of: $a + b$ and represent the result in single precision floating point format.

$a = 11.100_{\text{two}}$ and $b = 1.011_{\text{two}}$

CS-B: 6 Points

6

step 2: $a + b = 11.100 + 1.011 = 100.111$

step 2: Normalize

$$1.00111 \times 2^2$$

step 3: calc. the exponent

$$\Rightarrow 2 + 127 = 129$$

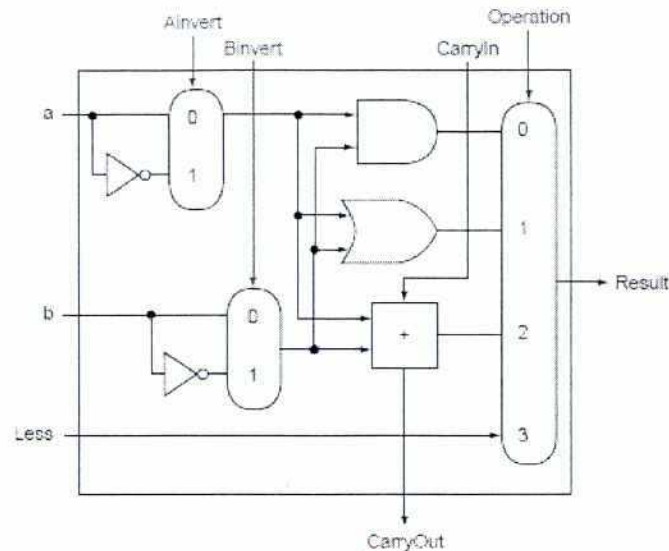
$$(0 \ 10000001 \ 0011100 \dots 00)_2$$

64

128

✓

5. A digital ALU that can execute some arithmetic and logical operations is shown below.



a. Show how to execute the **subtract** operation: $a - b$.

CS-A: 3 Points

3

$A_{invert} = 0_2$
 $B_{invert} = 1_2$
 $CarryIn = 1_2$
 $operation = 10_2$
~~Less = 0~~

a will go to the adder with no changes but we'll take the 1's complement of b and pass it to the adder, the adder will add $a + b$ then it'll add it with the carryIn which it's 1_2 to take the two's complement of the result.

b. Can you execute **NOR** operation on this ALU? Explain your answer.

CS-B: 3 Points

3

Yes, we can take the complement of a, b and then passing them into AND gate.

$A_{invert} = 1_2$
 $B_{invert} = 1_2$
 $CarryIn = 0_2$
 $operation = 00_2$
 $Less = 0$

check this

a	b	a'	b'	(a+b)	(a+b)'	a'.b'
0	0	1	1	0	1	1
0	1	1	0	1	0	0
1	0	0	1	1	0	0
1	1	0	0	0	1	0

$$a \text{ NOR } b \equiv (a + b)' \equiv a' \cdot b'$$

c. What are the necessary modifications in the given design if a new operation is added?

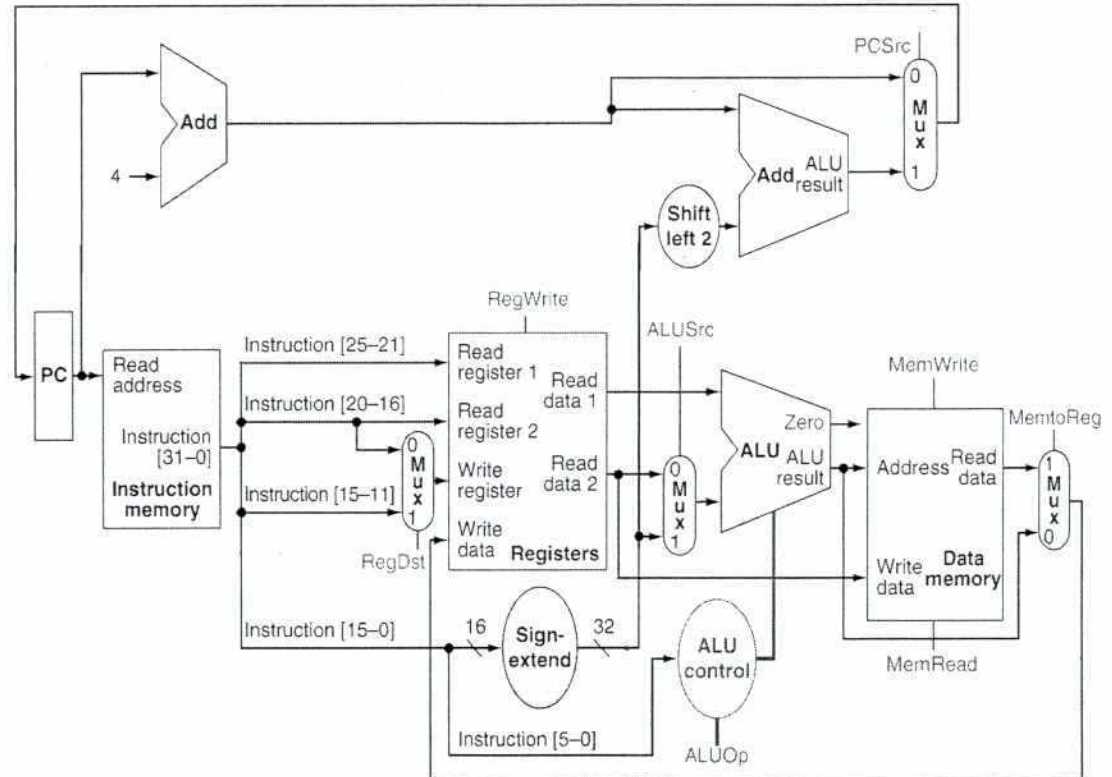
CS-C: 2 Points

2

if I considered the new operation inside chip or circuit such as the adder circuits.

generally we'll modify the mux ^{an input} by adding a new line into it and increasing its selection line by 1 extra bit, so we can represent number $(100)_2$ which it'll represent number 4 in decimal, which it's the ^{new} input in the mux for the new operation.

6. The next figure shows a datapath for a certain processor's architecture used for executing instructions.



a. What is the use of PC?

it contains the address of the instruction

that it's being executing. (it's used for determine which instruction should be executed)

CS-A: 2 Points

2

b. Why the design uses two memories?

one for storing the instructions and the other

one is for storing/loading data. the IM is small and faster than

Data memory which it'll take ^{more} time if we'll go to Data memory to load

c. List the steps of instruction fetch.

- get the address from PC then add 4 to it using the adder.
- look for the address in the IM then fetch the instruction in that address.

CS-A: 2 Points

1

d. What is the use of the shift left 2 unit.

to multiply the value of PC after adding

4 to it by 2, it's used in the branch instructions for calculate the label addresses and to jump to it.

CS-A: 2 Points

1/2

e. List the steps of writing the ALU result to a destination register.

ALU → mux (where the input of MemtoReg = 0)

→ to Register file (where RegWrite should be 1)

CS-A: 2 Points

1/2

